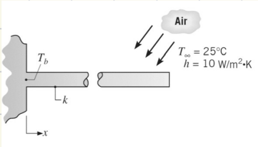
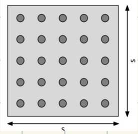
Heat transfer HW 5

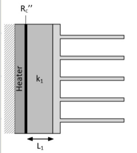
1) • **known**: cylindrical pin fin, $r = 1\,mm$, $L = 5\,cm$, $T_\infty = 25°C$, $h = 10\ W/(m^2 K)$, $k = 200\ W/m\cdot K$

• assumptions: thermally insulated, steady state

• schematic:







• find: a) fin temp @ $x = L/2$, $x = L$ ($T_b = 125°C$)

b) ε_f, η_f (3.4)

c) η_f (3.5), compare w/ b

d) R_f

e) 5x5 array, $s = 3cm$. η_o & R_o. Heat dissipated

f) thermal resistance network, T_b.

• analysis:

a) $$\frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{\cosh[m(L-x)]}{\cosh(mL)}$$

$$m = \sqrt{\frac{hP}{kA_c}}$$ $A_c = \pi r^2 = \pi(0.001)^2 = 3.14\times10^{-6} m^2$

$P = 2\pi r = 2\pi(0.001) = 0.006283\ m$

$$m = \sqrt{\frac{10(0.006283)}{200(3.14\times10^{-6})}} = \sqrt{100} = 10\,m^{-1}$$ $mL = 10(0.05) = 0.5$

$\theta_b = T_b - T_\infty = 125 - 25 = 100°C$

$$\theta(L) = \frac{\theta_b}{\cosh(mL)} = \frac{100}{1.127} = 88.7$$

$T(L) = 25 + 88.7 = \boxed{113.7°C = T(L)}$

$x = L/2 = 0.025$ $m(L-x) = 10(0.05 - 0.025) = 0.25$

$\cosh(0.25) = 1.031$

$\theta(L/2) = 100\left(\frac{1.031}{1.127}\right) = 91.5$

$T(L/2) = 25 + 91.5 = \boxed{116.5°C = T(L/2)}$

b) $q_f = M\tanh(mL)$

$M = \sqrt{hPkA_c}\;\theta_b = \sqrt{(10)(0.006283)(200)(3.14\times10^{-6})}\;(100)$

$M = 0.628$

$\tanh(0.5) = 0.462$

$q_f = (0.628)(0.462) = \boxed{0.290\ W = q_f}$

$\varepsilon_f = \dfrac{q_f}{hA_b(T_b - T_\infty)}$ $\quad A_b = A_c = 3.14\times10^{-6}$

$\varepsilon_f = \dfrac{0.290\ W}{(10)(3.14\times10^{-6})(100)} = \boxed{92.4 = \varepsilon_f}$

$\eta_f = \dfrac{q_f}{hPL(T_b - T_\infty)} = \dfrac{0.290}{(10)(0.006283)(0.05)(100)} = \dfrac{0.290}{0.314}$

$\boxed{\eta_f = 0.924}$

c) $\eta_f = \dfrac{\tanh(mL_c)}{mL_c} = \dfrac{\tanh\left(10\cdot\left(0.05+\frac{0.002}{4}\right)\right)}{(10)\left(0.05+\frac{0.002}{4}\right)}$

$\boxed{\eta_f = 0.923}$

0.1% diff

d) $R_f = \dfrac{T_b - T_\infty}{q_f} = \dfrac{100}{0.290} = \boxed{344.8\ K/W = R_f}$

e) $s = 0.03$ $\quad A = (0.03)^2 = (9\times10^{-4})\ m^2$

$A_f = \pi DL = \pi(0.002)(0.05) = 3.14\times10^{-4}\ m^2$

$A_{f,total} = 25A_f = 0.00785\ m^2$

$A_{total} = (9\times10^{-4}) - (7.85\times10^{-5}) + 0.00785 = 0.00867\ m^2$

$\eta_o = 1 - \dfrac{A_{f,tot}}{A_{tot}}(1-\eta_f) = 1 - \dfrac{0.00785}{0.00867}(1-0.924)$

$\boxed{\eta_o = 0.931}$

$q_{fins} = 25(0.290) = 7.25\ W$

$q_{base} = h A_{unfinned} (T_b - T_\infty)$

$= (10)(9\times10^{-4} - 7.85\times10^{-5})(100) = 8.07\ W$

$R_o = \frac{T_b - T_\infty}{q_{tot}} = \frac{100}{8.07}$

$\boxed{R_o = 12.4\ K/W}$

f) $q'' = 1000\ W/m^2$ $\quad A = 9\times10^{-4}$

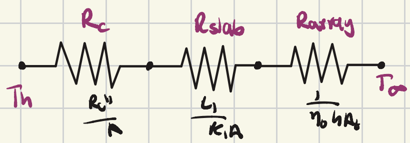
$q = q''A = (1000)(9\times10^{-4}) = 0.9\ W$

$R_c = \frac{10^{-4}}{(9\times10^{-4})} = 0.111$

$R_{slab} = L_1/k_1A = (0.02)/(10)(9\times10^{-4}) = 2.22$

$R_{total} = 0.111 + 2.22 + 12.4 = 14.73$

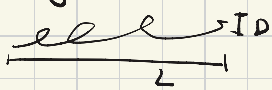
$T_h = T_\infty + q R_{tot} = 25 + 0.9(14.73) = 38.3°C$



2) • **known**: $D = 1\ mm$, $T_\infty = 20°C$, $\rho = 4000\ kg/m^3$, $k = 400\ W/m\cdot K$, $c = 500\ J/kg\ K$
$R_e' = 0.01\ \Omega/m$, $h = 500\ W/m^2K$, $t = 0\ s$, $I = 100\ A$

• **assumptions**: steady state

• **schematic**



• **find**

a) steady state temp

b) time it takes for wire to reach a temp that is within 5°C of steady-state temp

• **analysis**:

a) $q' = I^2 R_e' = (100^2)(0.01) = 100\ W/m$

$q' = hP(T_s - T_\infty)$

$(\pi)(0.001)(500)(T_s - 20) = 100\ W/m$

$100 = 1.57(T_s - 20) \Rightarrow T_s - 20 = 63.7$

$\boxed{T_s = 83.7°C}$

b) $T(t) - T_\infty = (T_{ss} - T_\infty)(1 - e^{-t/\tau})$

$$\tau = \frac{\rho c V}{hA} = \frac{\rho c(\pi D^2/4)}{h(\pi D)} = \frac{\rho c D}{4h}$$

$$\tau = \frac{(4000)(500)(0.0011)}{4(500)} = \boxed{1s = \tau}$$

$T = 83.7 - 5 = 78.7$

$78.7 - 20 = (63.7)(1 - e^{-t}) \Rightarrow 58.7 = 63.7(1 - e^{-t})$

$e^{-t} = 0.079$

$$\boxed{t = 2.54s}$$

Homework 4 for Heat Transfer taught me a lot about thermal resistance networks. I think this is helpful to understand for the future because it creates a visual representation of the thermal resistances and temperatures of the system we are solving, making it easier to understand the process as a whole. Additionally, this homework was essential for my understanding of square fins, how they work, and how to solve for temperatures regarding them because before this assignment the concept was not quite clear. This is important because pin fins will be important in my career as a mechanical engineer.